Application of Monte Carlo Sampling to Calculate Quantum Binding Free Energies from DFT Total Energies

Chris Sampson

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1 Introduction

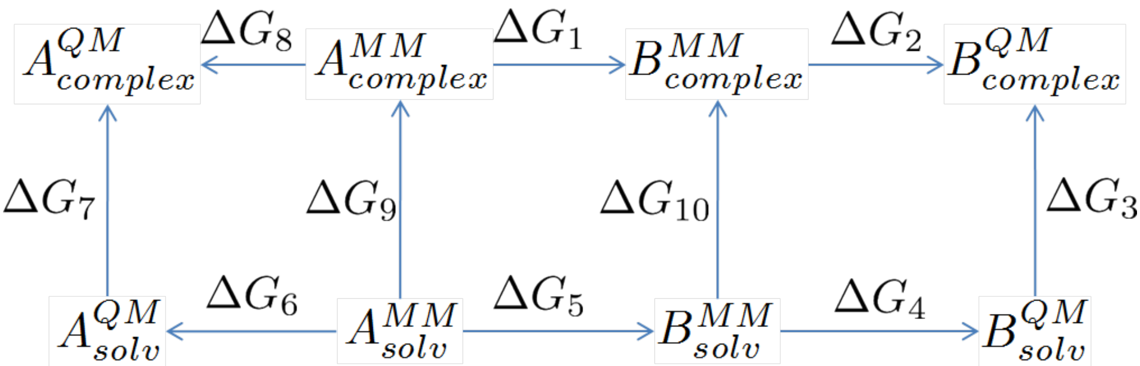

Figure 1: Extended Thermodynamic Cycle

$$\Delta G = -k_B T \ln \left\langle \exp\left[-\beta \Delta U_{QM-MM}\right] \right\rangle_{MM}. \tag{1}$$

$$U_{interaction} = U_{complex} - U_{ligand} - U_{host}. \tag{2}$$

PROGRESSION FROM PREVIOUS CHAPTER, HOW IT IMPROVES UPON IT ETC...

2 Theory

The method to generate an accurate QM/MM ensemble can be simply described by a nested loop. This section will be split into sections, first the description of the inner loop (generating the classical ensemble), then the acceptance to a QM/MM ensemble, then the perturbation to the full QM description and finally simulation details.

2.1 Hybrid Monte Carlo

In order to apply the Metropolis-Hastings criterion when peturbing from MM to QM/MM, the generation of MM structures must obey the detailed balance condition (equation 3). This can be satisfied by using pure Monte Carlo or by Hybrid Monte Carlo [1]. By using Hybrid Monte Carlo, the random walk errors associated with Monte Carlo are avoided and the conformational movements between $\mathbf{R}$ and $\mathbf{R}'$ can be larger [2].

$$\rho(\mathbf{R})\pi(\mathbf{R} \to \mathbf{R}') = \rho(\mathbf{R}')\pi(\mathbf{R}' \to \mathbf{R}), \tag{3}$$

where $\rho(\mathbf{R})$ is the probability of occupying a certain conformation $\mathbf{R}$, from here onwards this will be referred to as state probability. $\pi(\mathbf{R} \to \mathbf{R}')$ is the transition probability shown in equation 4, which consists of the trial probability and the acceptance probability.

$$\pi(\mathbf{R} \to \mathbf{R}') = \pi_{acc}(\mathbf{R} \to \mathbf{R}')t(\mathbf{R} \to \mathbf{R}'), \tag{4}$$

$\pi_{acc}(\mathbf{R} \to \mathbf{R}')$ is the acceptance probability, given by a Metropolis-Hastings criterion, and can be seen in equation 5, and $t(\mathbf{R} \to \mathbf{R}')$ is the trial probability, which is the probability of moving to a new conformation.

$$\pi_{acc}(\mathbf{R} \to \mathbf{R}') = min\left\{1, \frac{\rho(\mathbf{R}')t(\mathbf{R}' \to \mathbf{R})}{\rho(\mathbf{R})t(\mathbf{R} \to \mathbf{R}')}\right\} \tag{5}$$

The generation of new conformations in hybrid Monte Carlo is performed by an underlying MD simulation. In order to satisfy detailed balance the MD must be in the microcanonical ensemble as interaction with a thermostat will lead to a non-reversible path. Two inputs are required each time a simulation is performed in the microcanonical ensemble, these are the structure and the velocities. The acceptance probability shown in equation 5 requires two inputs and their reverse values. One is the state probability of both the original conformation (the reverse in this case is the state probability of the new conformation) and the trial probability. The state probability is given by the following Boltzmann distribution,

$$\rho(\mathbf{R}) = \frac{\exp(-\beta U(\mathbf{R}))}{Z(N, V, T)}. \tag{6}$$

Where $Z(N, V, T)$ is the configurational partition function.

The trial probability required for the acceptance criterion is related to the selected momenta. To ensure a Boltzmann distribution of kinetic energy, the momenta is randomly selected from a gaussian distribution using the Marsaglia Polar method [3]. By selecting momenta like from a gaussian, this controls the temperature of the simulation, thus acting as a thermostat.

$$t(\mathbf{R} \to \mathbf{R}') \propto \prod_{i=1}^{n} \frac{1}{\sqrt{2m_i \pi k_B T}} \exp\left[-\beta \frac{\mathbf{p}_i^2}{2m_i}\right] \tag{7}$$

To derive a useful form of the acceptance criterion placing equation 6 and 7 into equation 5, noticing that equation 7 is the functional form of kinetic energy, the following is true,

$$\begin{aligned}
\pi_{acc}(\mathbf{R} \to \mathbf{R}') &= min\left\{1, \frac{\exp(-\beta U(\mathbf{R}')) \exp(-\beta K(\mathbf{P}'))}{\exp(-\beta U(\mathbf{R})) \exp(-\beta K(\mathbf{P}))}\right\} \\
&= min\left\{1, \frac{\exp(-\beta H(\mathbf{R}', \mathbf{P}'))}{\exp(-\beta H(\mathbf{R}, \mathbf{P}))}\right\} \\
&= min\left\{1, \exp(-\beta(H(\mathbf{R}', \mathbf{P}') - H(\mathbf{R}, \mathbf{P})))\right\} \\
&= min\left\{1, \exp(-\beta \Delta H(\mathbf{R}, \mathbf{P}))\right\}.
\end{aligned} \tag{8}$$

By running the MD simulation in the microcanonical ensemble the total Hamiltonian energy should be conserved and as such each snapshot will be accepted, these snapshots are then accepted into the canonical ensemble. Any variation from the total Hamiltonian energy will alter the classical acceptance ratio, but will only have occured due to errors in the integrator, these can be minimised by altering the timestep i.e. the smaller the timestep the better energy conservation. This method will produce an accurate classical canonical ensemble of structures.

After n_{MaxMM} Hybrid Monte Carlo steps the QM/MM energy of the system is calculated and a Metropolis-Hastings criterion is applied.

2.2 Accepting to the QM/MM ensemble

The QM/MM model used within this method is the ONIOM approach (mechanical embedding)[4]. This is provides a simple QM/MM description of the system by the following,

$$U^{QM/MM}_{complex} = U^{MM}_{complex} - U^{MM}_{region} + U^{QM}_{region}. \tag{9}$$

Where $U^{MM}_{complex}$ is the total complex energy of the system when calculated using a classical potential and U^{QM}_{region} is the quantum potential energy of the region of interest (e.g. a binding pocket and ligand).

Using this description of QM/MM, every n_{MaxMM} Hybrid Monte Carlo steps the QM/MM energy of the last accepted classical structure is calculated. This energy is then placed in an additional Metropolis Hastings criterion, equation 5. To derive the terms required for the criterion, $\rho(\mathbf{R})$ refers to the state probability of the QM/MM structure, and is given by $U^{QM/MM}_{complex}$. The trial probability required can be derived by the following [5],

$$\begin{aligned} \frac{t(\mathbf{R}_n, \mathbf{R}_{n-1}, \mathbf{R}_{n-2}, \ldots, \mathbf{R}_0)}{t(\mathbf{R}_0, \mathbf{R}_1, \mathbf{R}_2, \ldots, \mathbf{R}_n)} &= \frac{\pi(\mathbf{R}_1 \rightarrow \mathbf{R}_0)}{\pi(\mathbf{R}_0 \rightarrow \mathbf{R}_1)} \cdots \\ &\quad \frac{\pi(\mathbf{R}_n \rightarrow \mathbf{R}_{n-1})}{\pi(\mathbf{R}_{n-1} \rightarrow \mathbf{R}_n)}. \end{aligned} \tag{10}$$

Then, by applying the detailed balance condition equation 3,

$$\frac{t(\mathbf{R}_n, \mathbf{R}_{n-1}, \mathbf{R}_{n-2}, \ldots, \mathbf{R}_0)}{t(\mathbf{R}_0, \mathbf{R}_1, \mathbf{R}_2, \ldots, \mathbf{R}_n)} = \frac{\rho(\mathbf{R}_0)}{\rho(\mathbf{R}_1)} \cdots \frac{\rho(\mathbf{R}_{n-1})}{\rho(\mathbf{R}_n)}. \tag{11}$$

This relates the QM/MM trial probability to the classical transition probability, and by cancellation the following is obtained,

$$\frac{t(\mathbf{R}_n, \mathbf{R}_{n-1}, \mathbf{R}_{n-2}, \ldots, \mathbf{R}_0)}{t(\mathbf{R}_0, \mathbf{R}_1, \mathbf{R}_2, \ldots, \mathbf{R}_n)} = \frac{\rho(\mathbf{R}_0)}{\rho(\mathbf{R}_n)}. \tag{12}$$

Leading to the acceptance criterion of the following,

$$\pi_{acc}(\mathbf{R} \rightarrow \mathbf{R}') = min\left\{1, \exp\left[-\beta(\Delta\Delta U)\right]\right\}, \tag{13}$$

where,

$$
\begin{aligned}
\Delta\Delta U = \quad & [U_{QM/MM}(j;\lambda) - U_{MM}(j)] \\
- \quad & [U_{QM/MM}(i;\lambda) - U_{MM}(i)].
\end{aligned} \tag{14}
$$

This will produce an accurate QM/MM ensemble of structures at 300K.

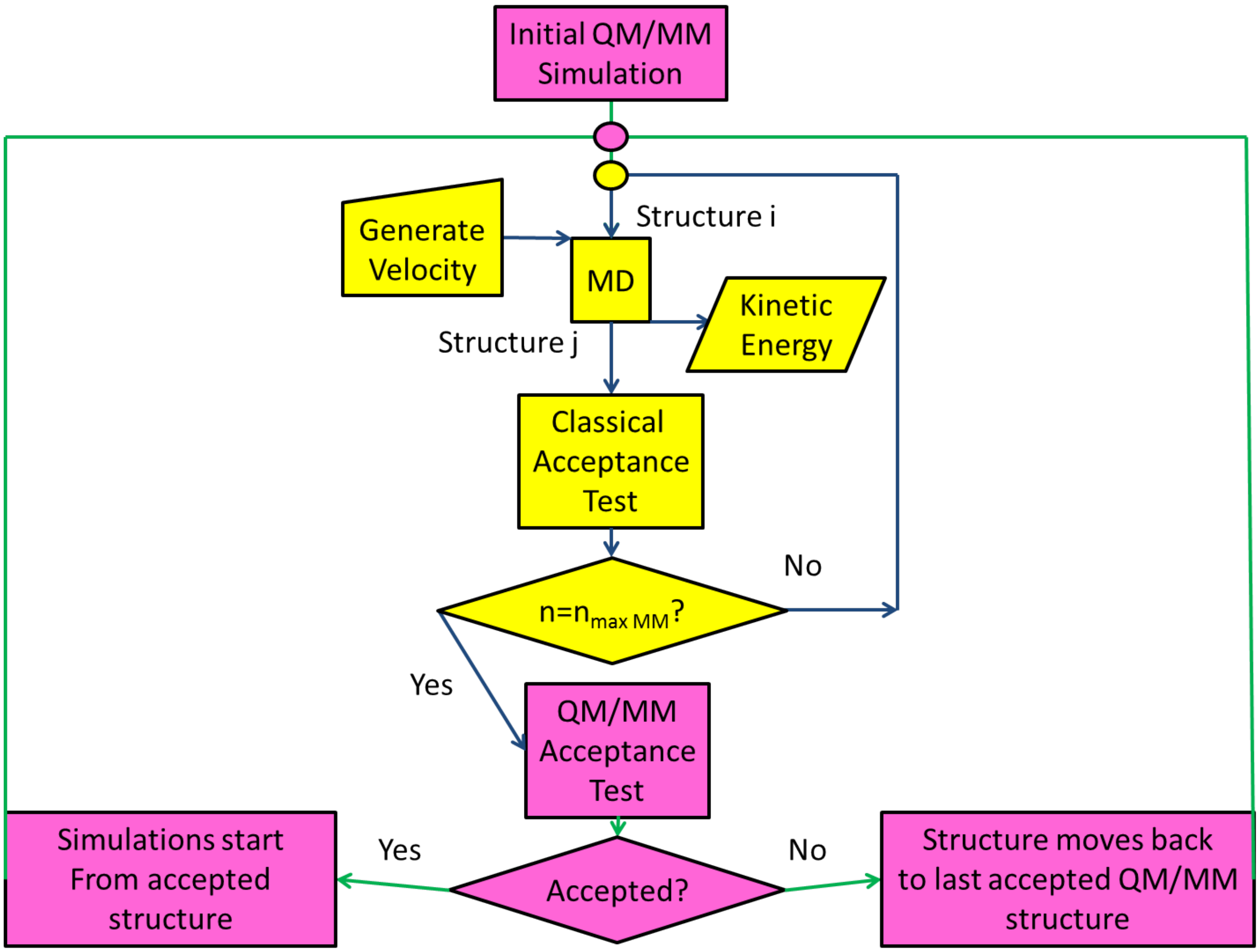

Figure 2: A flow chart describing how the method accepts to the QM/MM level, the inner loop (yellow and blue) represents hybrid Monte Carlo, the outer loop (pink and green) is the QM/MM level

2.3 Stratification

In order to make the transition from MM to QM/MM smoothly additional λ steps can be introduced. To do this a simulation is performed in an identical fashion as previously described, with the only exception within the $\Delta\Delta U$ of equation 14. Where $U_{QM/MM}(j;\lambda)$ becomes $U_{QM/MM}(\lambda)$. And $U_{QM/MM}(\lambda) = (1-\lambda)U_{classical} + (\lambda)U_{QM/MM}$.

The free energy is then obtained using thermodynamic integration (TI),

$$
\begin{aligned}
\Delta G &= \int_0^1 d\lambda \left\langle \frac{\partial U(\lambda)}{\partial \lambda} \right\rangle_\lambda \\
\frac{\partial U(\lambda)}{\partial \lambda} &= U_{QM/MM} - U_{classical}
\end{aligned}
$$

$$\Delta G = \int_0^1 d\lambda \left\langle U_{QM/MM} - U_{classical} \right\rangle_\lambda$$

2.4 Transitioning to the full QM

By making the assumption that the $U_{QM/MM}$ ensemble is an accurate representation of the U_{QM} ensemble, the transition from QM/MM to QM is simple. To perform this perturbation, the energy differences $U_{QM} - U_{QM/MM}$ are calculated then the Zwanzig equation can be used in a similar fashion to equation 1.

3 Methods

3.1 Classical details

REWRITE SECTION WHEN ALL RESULTS ARE IN!

All parameters for non-waters, unless otherwise specified, were obtained using antechamber [6]. The equilibration procedure was undertaken within the single precision version of Gromacs v4.6.2 [7] and is as follows. For ethane and ethanol in 448 and 450 waters respectively, leapfrog integrator was used to initially minimised with the steepest descent algorithm for 5000 steps with a cutoff of 11 Å, electrostatics were treated with PME and a cut-off method was used for Van der Waals interactions. Following this a 500 ps simulation in the canonical ensemble was run using a timestep of 1 fs, the Berendsen thermostat was used to heat the system smoothly to 300 K, finally a 1ns isothermal-isobaric simulation was performed using the Berendsen barostat. After equilibration the box size decreased sufficiently to only allow a maximum of 8 Å cut-off for the non-bonded interactions.

The water model used was a flexible SPC model described by Wu et al. [8]. This was used as it provided better energy conservation within the microcanonical simulations and it was thought it would represent a quantum ensemble more accurately.

PUT IN TABLE OF PARAMETERS!!

The process for N_2 in vacuum was similar, although the pressure equilibration was skipped.

The classical simulation detail within the hybrid Monte Carlo were performed using the velocity-verlet algorithm, with a 0.25 fs timestep, the reason for such a small timestep can be seen in figure 3. Which shows the affect of changing the timestep for an MD simulation run 3 times. The overall energy is conserved, but the fluctuations cause a lower classical acceptance rate when any larger timestep is used. Each simulation is run for 20 ps and for the ethane and ethanol in water systems, 200 steps of pure hybrid Monte Carlo was performed as equilibration, following this only a single classical simulation is run between QM/MM simulations.

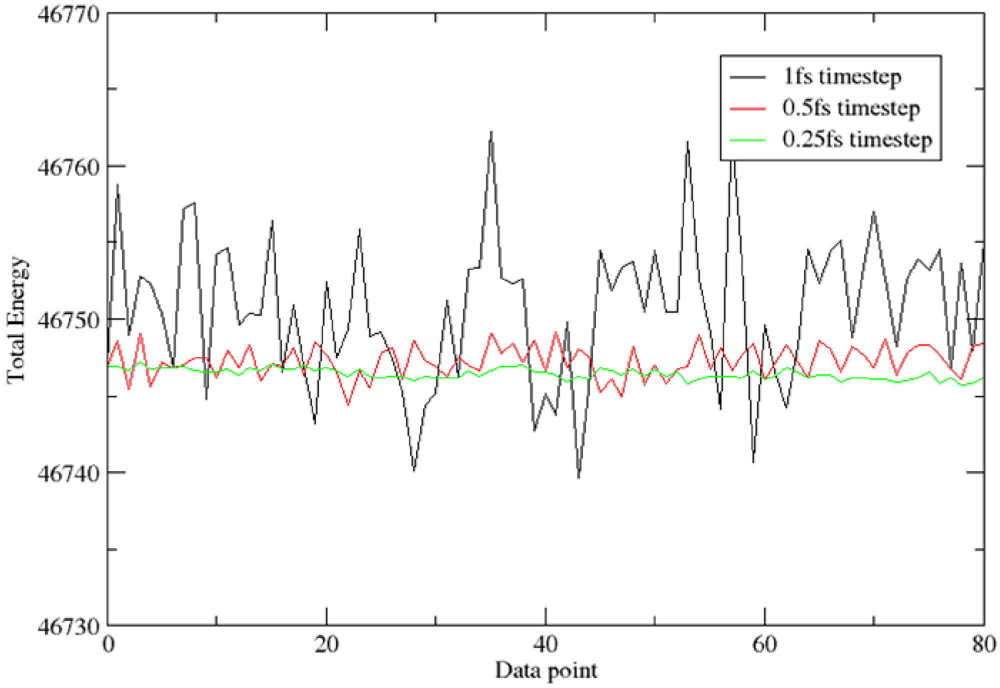

Figure 3: Running an MD simulation using different timesteps to show fluctuations of the total energy in the micro-canonical ensemble

3.2 QM details

The Quantum energies were calculated with 2 pieces of software, for testing NWCHEM [9] was used, then in order to move to fully Quantum description ONETEP [10] was then used. In both cases 100 steps of equilibration was initially performed, this includes 100 applications of the acceptance test on the QM/MM and MM energies.

When using NWCHEM, the exchange-correlation functionals of pbe96 were used with a cc-pvtz basis set for all atoms. The setup for ONETEP, in the case of N_2, was that 4 NGWFs were used each being 7 bohr. For ethane and ethanol 4 NGWFs were used on heavy atoms and 1 for light atoms, all were 8 bohr and in both cases a kinetic energy cutoff of 800eV was used.

TALK ABOUT DISPERSION MODEL

4 Results

4.1 A Comparison of Methods While modifying the Force Field

A simple test system of N_2 in vacuum was selected to show the accuracy of the generated QM ensemble when comparing the approach detailed within this article, applying the Metropolis-Hastings criterion, to the direct method of extracting snapshots from an MD trajectory at regular intervals and applying a single step free energy perturbation to the QM level. The advantage of such a simple test system is that we cancel out any need for a classical region, meaning that we accept into a purely quantum ensemble.

For simplicity, the snapshots required for the single step perturbation approach (SSP) were taken from the ensemble when λ=0 of the Metropolis-Hastings applied method (MHA). These snapshots more accurately represent the classical ensemble than if a simulation had been run in the canonical ensemble as we minimise errors within the integrator.

All parameters were obtained for N_2 from antechamber [6] with the exception of the equilibrium bond length which was set to the experimental length of 1.098 [11]. The bond length within the force field was then increased in increments of 0.01 Å and the affect on the proposed quantum ensemble was observed. The results can be seen in table 1.

1.098	SSP	MHA	
+	Average Bond Length	Average Bond Length	Acceptance at λ=1
0.00	1.104	1.104	61.2
0.01	1.110	1.104	64
0.02	1.120	1.105	50.4
0.03	1.130	1.103	35.6
0.04	1.138	1.110	28.4
0.05	1.149	1.113	16.6
0.06	1.159	1.100	7.0
0.07	1.169	1.100	2.2
0.08	1.179	N/A	0.0
0.09	1.188	1.117	1.0
0.10	1.197	N/A	0.0

Table 1: The affect on the average bond length (shown in Å) of the quantum ensemble when applying SSP and MHA and the acceptance rate (%) of MHA at λ=1

As expected, taking snapshots directly from an MD trajectory when the force field is parameterised badly, produces a poor representation of a quantum ensemble, which can be seen by examining the average bond lengths of table 1. However, when structures have to go through an acceptance criterion, as in MHA, the average bond length is conserved close to the experimental value. By forcing the classical and quantum configurational space to overlap less, the acceptance at λ=1 is drastically effected, in some cases no structures were accepted post-equilibration. This shows the acceptance rate is a good way to gauge how well the force field represents the quantum region: if the acceptance rate is very low then the force field is badly paramterised. In a simple system such as this, identifying which parameter is the main cause of the poor acceptance rate is trivial, however, when moving to larger systems this will be highly non-trivial. So, although the issue is identified the solution remains difficult to obtain.

The free energies of moving from the MM potential to the QM potential were then calculated and are shown in figure 1.

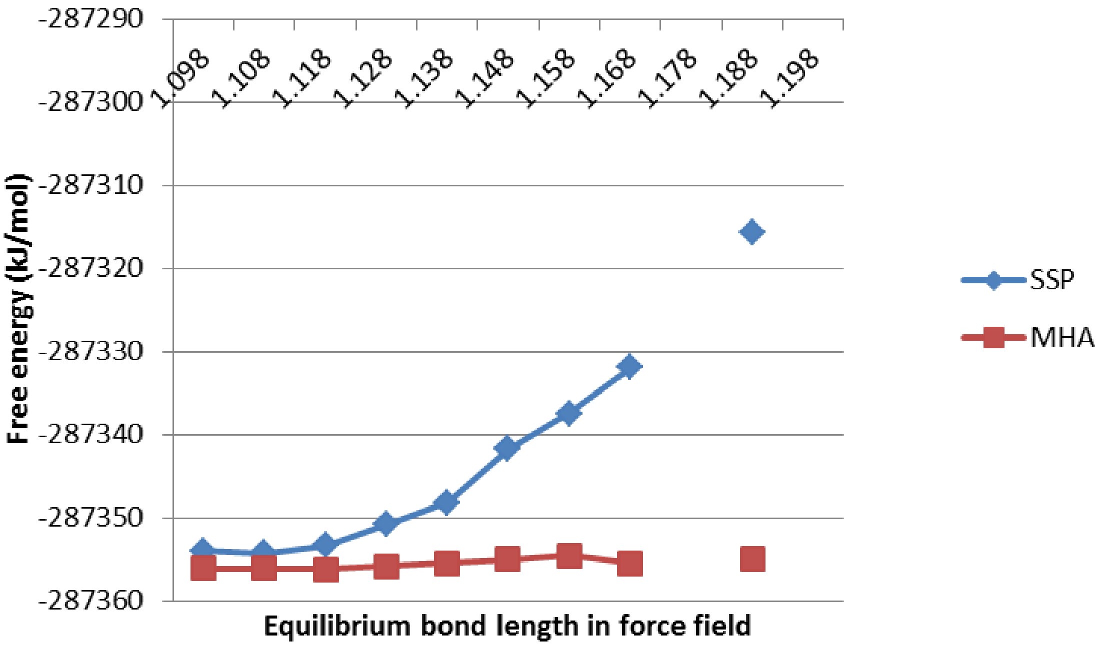

Figure 4: Free energy as a function of the bond length when moving from MM to QM

It is apparent by examination of figure 1 that the free energies diverge from each other as the equilibrium bond length is increased in the classical potential. The explanation for this is found when considering the potential of mean force (PMF) plots produced when using λ windows. These are shown in figure 5.

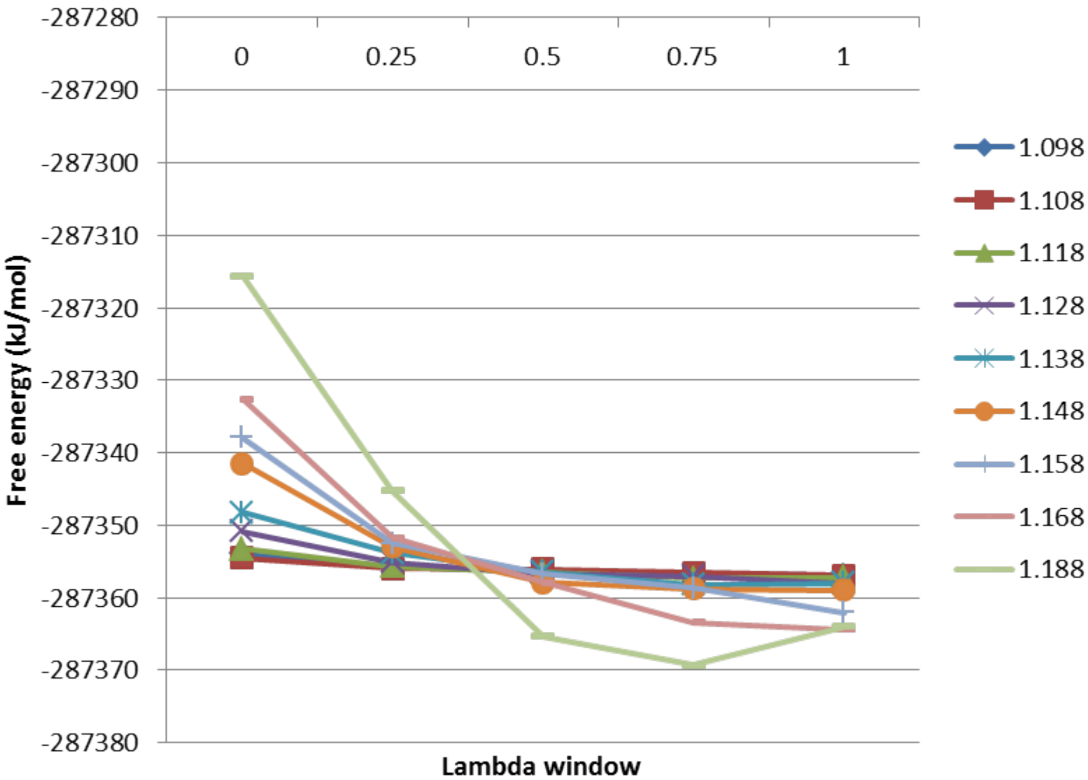

Figure 5: PMFs for the free energy as a function of the bond length when moving from MM to QM for MHA

The Zwanzig equation (equation 1) assumes the PMF will be a straight line between the MM ($\lambda = 0$) and QM ($\lambda = 1$). This assumption proves to be catastrophic to the free energies as the force field parameters move from the ideal values. These results provide us with confidence that the MHA method can improve upon results obtained using the SSP method. The next section applies these methods to more realistic test systems.

4.2 Generating QM/MM ensemble

After the promising results when using N_2, more realistic test systems were chosen to be ethane in 448 waters and ethanol in 450 waters. The introduction of water now provides us with the classical region within the ONIOM [4] approach, so only the ligand is treated quantum mechanically. For testing purposes, initially NWCHEM was used to calculate the quantum ligand. This was later changed to ONETEP so the transition to full QM could be performed. The NWCHEM results will be shown first followed by results obtained when using ONETEP, then the following section will cover the perturbation to the full QM.

To evaluate how converged the free energy between the classical and QM/MM ensembles are, A and B within figure 1 are both set to be the same, i.e. when using ethane in 448 waters both A and B are ethane in 448 waters. The alchemical mutation usually required to obtain the free energy between these systems is then equal to 0. Two simulations are then performed from MM to QM/MM (run 1 and run 2) and the free energy compared. First the results for ethane in 448 waters is shown.

	SSP	MHA	
	Free Energy	Free Energy	Acceptance at λ=1
Run 1	-209340.77	-209342.23	41.80
Run 2	-209340.83	-209342.42	44.80
Difference	-0.06	0.18	

Table 2: Free energy convergence moving from MM to QM/MM for ethane in 448 waters, free energies shown in kJ/mol and acceptance is shown as %

In this case, both methods prove to converge the free energy to within thermal error (k_BT). Figure 6 shows why both methods converge well. The PMFs are both almost straight lines, the fluctuation in the PMF of run 1 shows is representative of the stochastic nature of the MHA method.

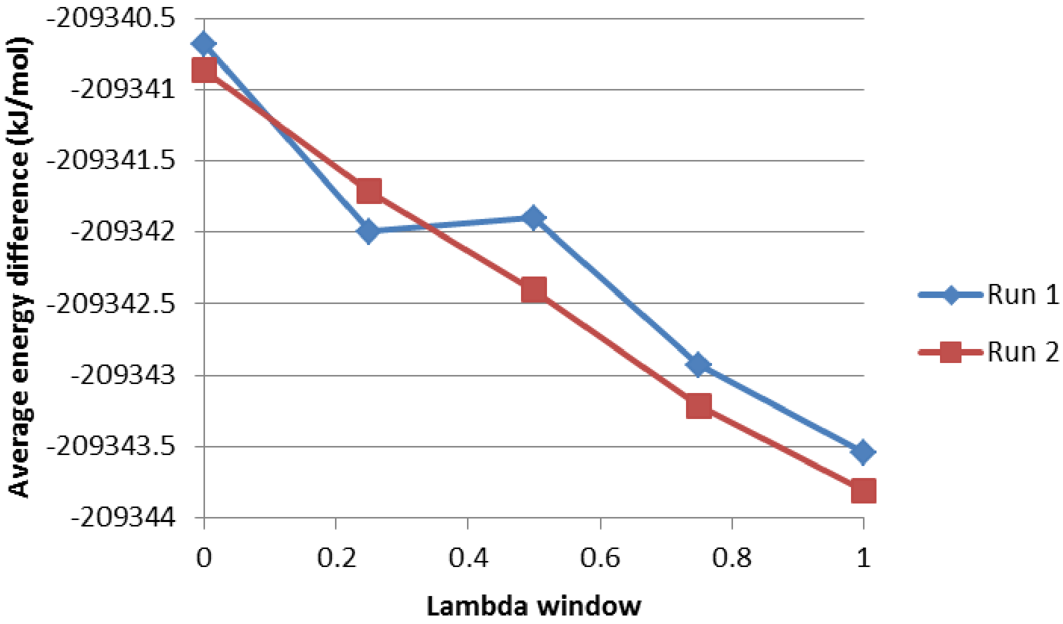

Figure 6: PMFs for the free energy for the two MHA runs for ethane in 448 waters using NWCHEM for the quantum region

To ensure convergence of the free energy shown for run 1, additional λ values were introduced at 0.18, 0.35, 0.6 and 0.88. The PMF was then plotted and is shown in figure 7

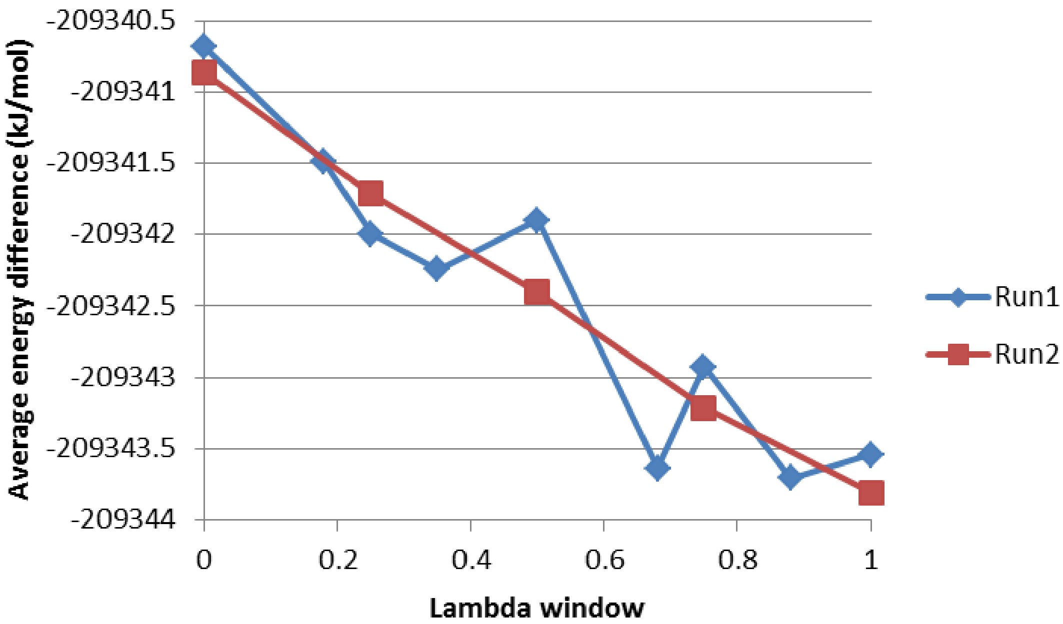

Figure 7: PMFs for the free energy for the two MHA runs for ethane in 448 waters using NWCHEM for the quantum region, with additional λ windows for run 1

The fluctuations within figure 7 run 1 again show the stochastic nature of the method. However, by introducing these additional λ steps the free energy for run 1 changes from -209342.23 kJ/mol to -209342.43 kJ/mol, which means the difference between run 1 and run 2 free energies is now 0.01 kJ/mol, proving convergence.

The method was then applied to ethanol in 450 waters and the results can be seen in table 3.

	SSP	MHA	
	Free Energy	Free Energy	Acceptance at λ=1
Run 1	-406688.60	-406693.31	16.80
Run 2	-406688.65	-406693.00	18.60
Difference	0.05	-0.31	

Table 3: Free energy convergence moving from MM to QM/MM for ethanol in 450 waters, free energies shown in kJ/mol and acceptance is shown as %

Both methods converge the free energy for ethanol in 450 waters, although interestingly the acceptance rates are considerably lower than those for ethane in 450 waters. The change in acceptance rate is attributed to the presence of the alcohol within ethanol, perhaps the OH bond vibration is modelled poorly by the classical region. The purpose of this study is to generate an accurate quantum ensemble, and although the acceptance rate is lower, the free energy is converged. A simulation was performed a $\lambda = 1$ which applies restraints on any bond that involves hydrogen. The resulting acceptance rate was 28.36%, which is an increase of approximately 10% when these bonds are allowed to vibrate. Although a higher acceptance, using flexible hydrogen bonds will be able to sample more structures, so these simulations will be used for analysis.

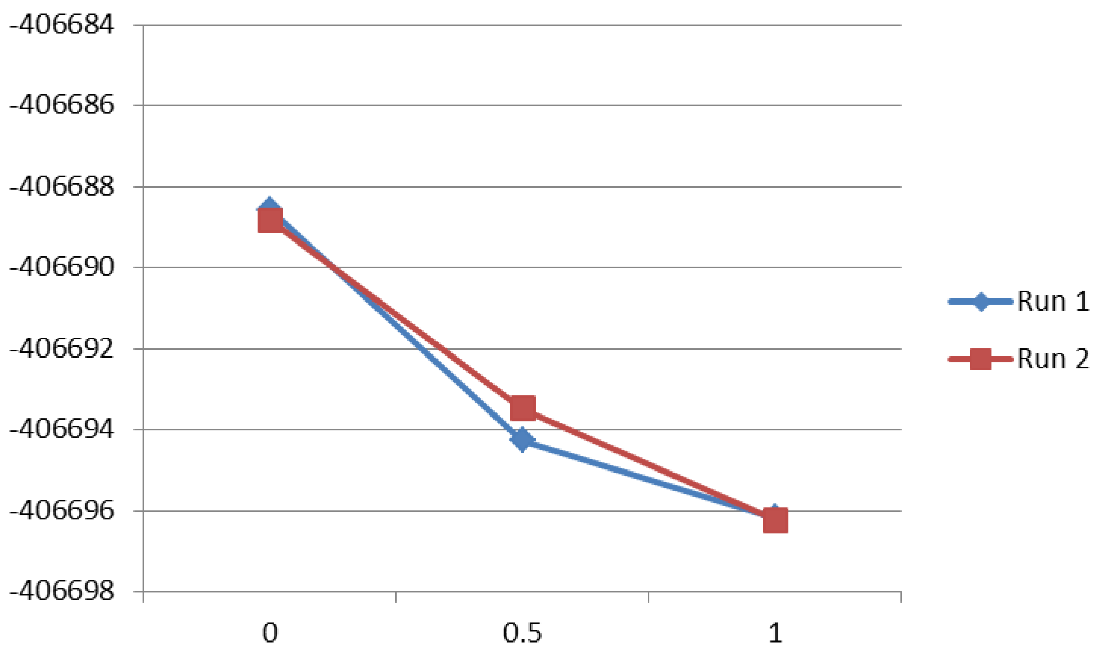

Figure 8: PMFs for the free energy for three MHA runs for ethanol in 450 waters using NWCHEM for the quantum region

The PMFs can be see in figure 8, which confirms how well converged the free energies are.

These results have shown that it is possible to go to a QM/MM level and converge the free energy, the next step is to perturb to the full QM ensemble. Because of this, the process was repeated using ONETEP. Only 3 λ windows were used with ONETEP, $\lambda = 0, 0.5, 1$. Figure 9 shows the obtained PMFs.

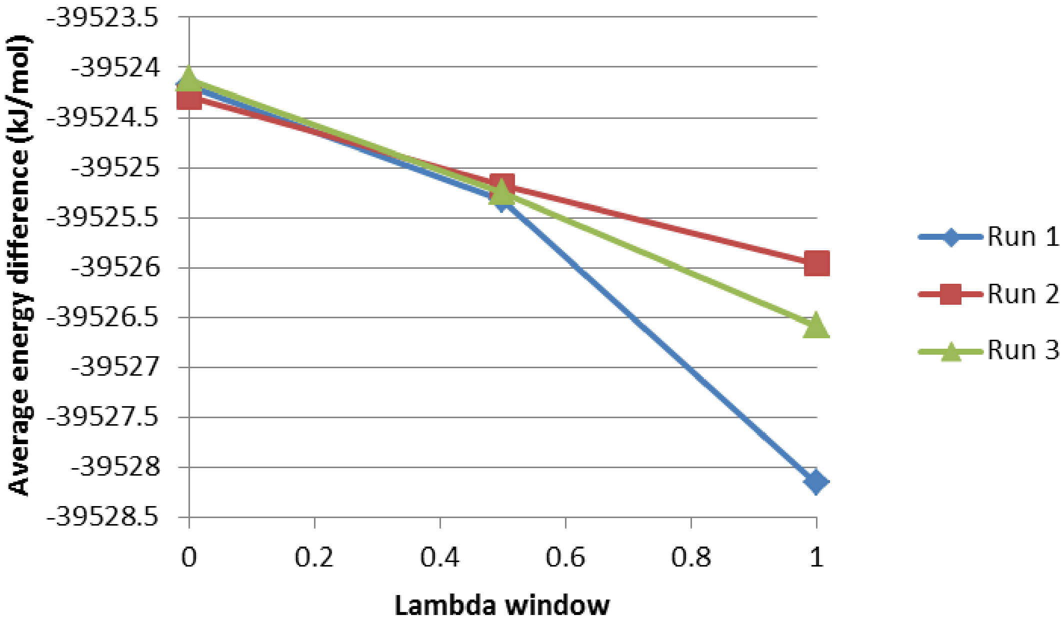

Figure 9: PMFs for the free energy for three MHA runs for ethane in 448 waters using ONETEP for the quantum region

Examination of figure 9 shows large differences within the average energy difference of $\lambda = 1$. This can be explained by considering the stochastic nature of the method. Even taking these differences to account the convergence of the free energies is largely unaffected (table 4) with the largest difference being just under 1 kJ/mol, within thermal error.

	SSP	MHA	
	Free Energy	Free Energy	Acceptance at λ=1
Run 1	-39524.18	-39525.18	34.09
Run 2	-39524.28	-39525.15	51.80
Run 3	-39524.19	-39524.19	43.00
Difference	0.10	0.99	

Table 4: Free energy convergence moving from MM to QM/MM, free energies shown in kJ/mol and acceptance is shown as % and the difference is the largest difference between the values

The difference in acceptance rates stems again from the stochastic nature of the method, in each case, however the free energy converges. The method was then applied to ethanol, the results can be see in table 5

	SSP	MHA	
	Free Energy	Free Energy	Acceptance at λ=1
Run 1	-81550.56	-81555.02	22.40
Run 2	-81551.09	-81555.44	21.0
Difference	0.53	0.42	

Table 5: Free energy convergence moving from MM to QM/MM for ethanol in 450 waters, free energies shown in kJ/mol and acceptance is shown as %

Free energies shown in table 5 are again converged with a lower acceptance rate than the acceptance rate for ethane. The PMFs can be seen in figure 10.

INSERT ETHANOL RESULTS WITH SHAKE!!!!

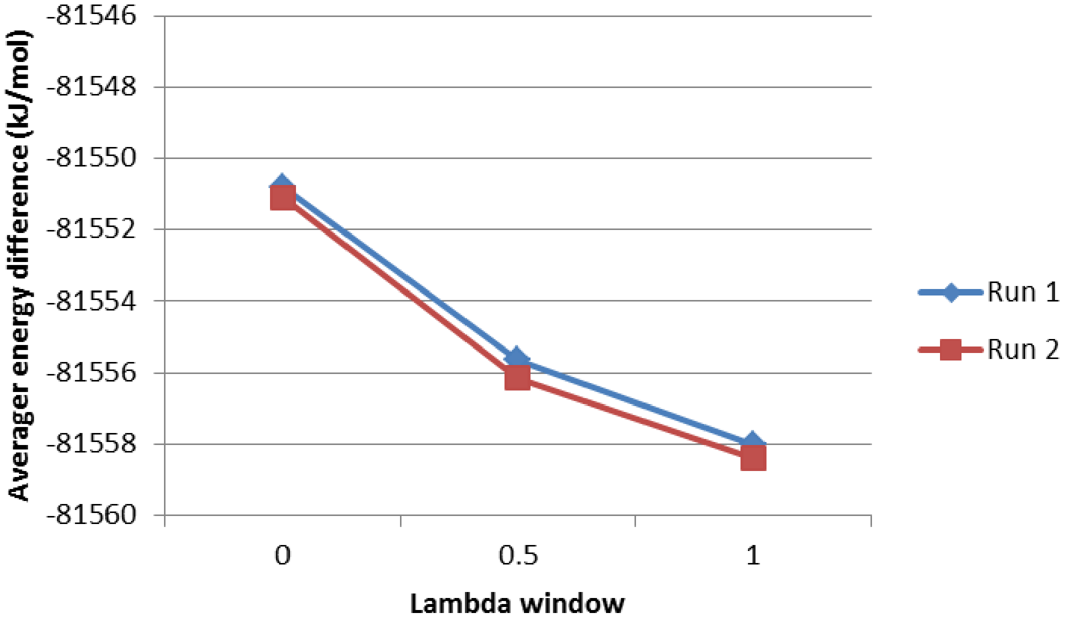

Figure 10: PMFs for the free energy for three MHA runs for ethanol in 450 waters using ONETEP for the quantum region

These results show good convergence, proving that moving from classical to QM/MM is possible using total energies. The next step is to move to the fully quantum ensemble.

4.3 Perturbation to full QM

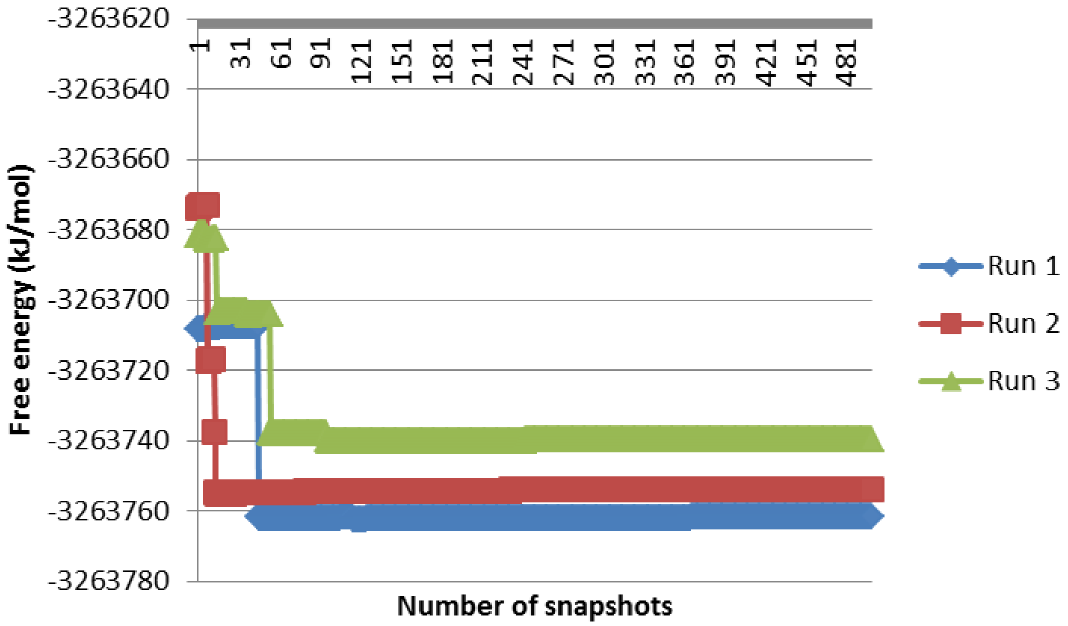

Figure 11: Free energy as a function of the number of snapshots when moving from the QM/MM to the QM for ethane in 448 waters

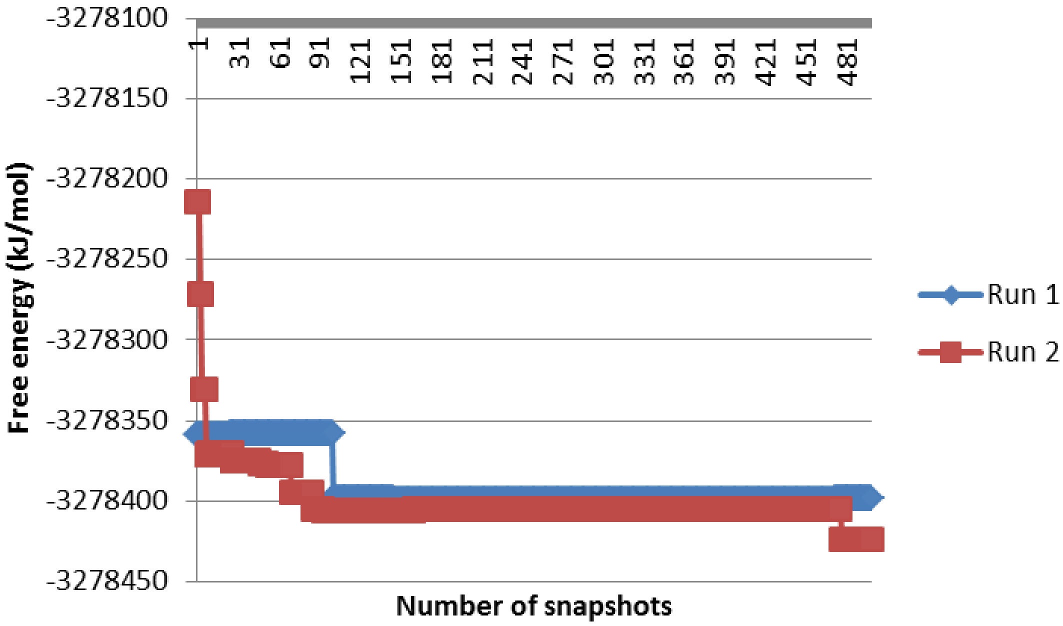

Figure 12: Free energy as a function of the number of snapshots when moving from the QM/MM to the QM for ethanol in 450 waters

Once the QM/MM ensembles of structures were built up, the next step was to perturb to the full QM, shown in figure 11 and figure 12. The "sawtooth" shape of the graph shows an issue with sampling [12]. The only approximation present within this method is that the QM/MM distribution is representative of the full QM ensemble. This approximation has proven to be false when considering only the ligand at QM level. In order to better represent a quantum ensemble, either the classical potential needs to be modified or the region of quantum space needs to be extended.

4.4 Extending the Quantum Region

In order to remove the sampling issues causing the "sawtooth" shape within figure 11 and 12 the quantum region within the QM/MM was extended to include water. To decide how many water to include the radial distribution function (RDF) was calculated for a 20ps simulation of ethane in 448 water and can be seen in figure 13.

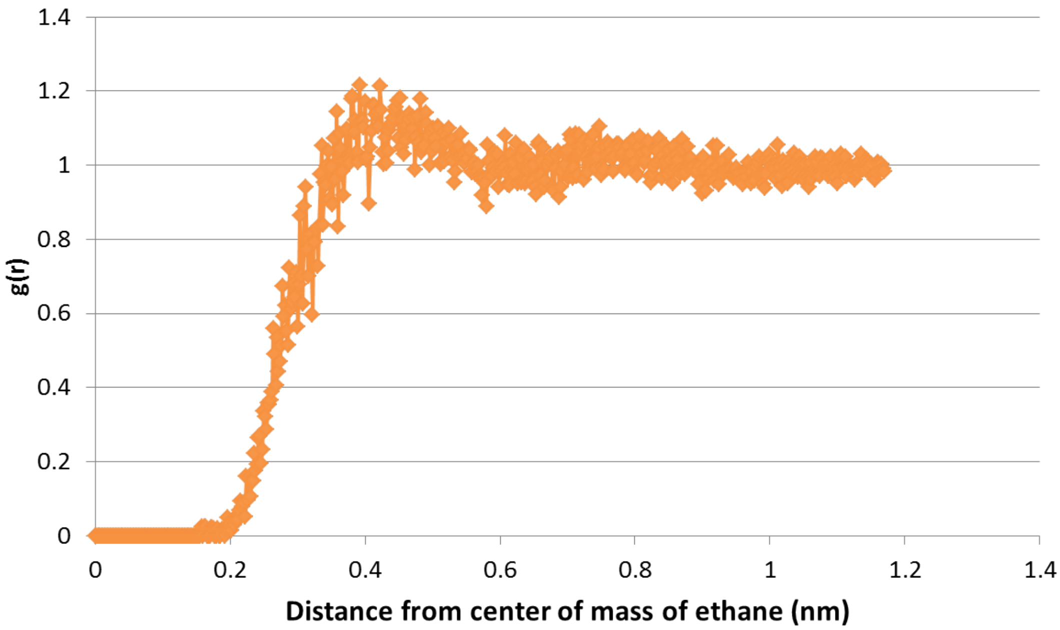

Figure 13: Radial Distribution Function for a 20 ps simulation of ethane in 448 water

As a first attempt to represent the QM configurational space better, the first solvation sphere was selected to be included within the quantum region. By examination of figure 13 it can be seen that the first solvation sphere ends approximately at 6 Å. The same simulation used to calculate the RDF was then used to calculate how many waters were within 6Å of ethane (using centre of mass). The result shows a maximum of 54 waters. The quantum region was then selected to be ethane and the closest 54 waters mechanically embedded within the remaining 394 waters.

The simulations (2 runs) were originally set up to obtain 600 quantum structures using ONETEP, as before, but after 380 both were cancelled. The reason for this is that after the 100 equilibration steps not a single structure was accepted, which is due to fluctuations of hundreds of kJ/mol within the calculated quantum energies.

The reason a set number of water was chosen, rather than a fixed cutoff was to avoid fluctuations due to different numbers of water within the quantum region. However, fluctuations are present regardless of this. The previous results show that the quantum energy fluctuation for ethane is within acceptable range (judging by the 40% acceptance rate). This leads to the conclusion that the fluctuations are either caused by the interaction between the water and ethane or the intramolecular energy of the water. As such each water within a snapshot was extracted and for speed the quantum energy was calculated using NWCHEM. The resulting energies had a range of XX kJ/mol showing a massive range of intramolecular energies. This fluctuation in energy shows that the flexible water model used isn't suitable to sample quantum conformational space.

ADD IN SENTANCE ABOUT THE IMPROVEMENT OF ACCEPTANCE OF ETHANOL

4.5 A simplification of the method

As stated in the theory section above, hybrid Monte Carlo allows the generation of an ensemble in a desired potential, that differs from the potential used to sample configurations. By considering this, it is possible to avoid accepting to a classical canonical ensemble, then moving to the QM/MM ensemble using two different acceptance tests; and building an ensemble on an ensemble. Instead, it is possible to change the acceptance criterion (equation 8) to include the QM/MM potential energy and λ windows could be introduced in the same manner as previously shown. This new implementation avoids the need for a nested loop and is conceptually simpler. This method can be considered by examining the inner loop (yellow and blue) of figure 2, where instead of the classical acceptance test you use the modified acceptance test, which makes the move to the QM/MM level redundant (in the outer loop). The inner loop is then cycled around until n=n_{maxMM}.

To validate the application of hybrid Monte Carlo (HMC) in this way, the test involving modifying the equilibrium bond length of N_2, performed in section 4.1 was repeated, this time using ONETEP for the QM calculations.

4.6 Applying HMC to N_2 while modifying the equilibrium bond length

1. N2 IN ARGON
2. N2 IN METHANE
3. N2 IN TIP3P
4. MM TO FULL QM USING HMC
5. MM EQUILIBRATION TO NVT THEN TO FULL QM
6. MM EQUILIBRATION TO ELEC EMBED
7. MM EQUILIBRATION TO ELECEMBED + LJ

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